

Figure 4.3d depicts how temporary decay can be determined from an exponential decay function. First, it is important to note the required properties of a decay function include: decreasing over time; always non-negative; approaches zero asymptotically, otherwise the decay function may include some permanent effect; and most important, provides an accurate representation of reality. Too often, we find quasi-quants selecting decay functions that possess only some of these required properties but when tested with real data the function does not provide an accurate description of the system. In many of these cases investors are better off using intuition than depending on insight from a faulty model.

A useful decay function that exhibits these properties and proves an accurate representation of reality is the exponential decay function. This function provides the percentage of temporary impact remaining over time (compared to $t = 0$) and is written as $d(t) = e^{-\gamma t}$. Here $\gamma > 0$ is the decay parameter that determines the rate of decay. Larger values of γ will decay at a faster rate than smaller values. From this expression the percentage of temporary impact that has already decayed at time t is: $1 - d(t)$.

An appealing property of the exponential decay function is that it decreases at a constant rate. In other words, the percentage reduction from one period to the next is the same. For example, with a parameter of $\gamma = 0.5$, the percentage of temporary impact remaining after the first period $d(1) = e^{-0.5 \cdot 1} = 0.6065$ and after two periods the percentage of temporary impact remaining is $d(2) = e^{-0.5 \cdot 2} = 0.3679$ and can also be written as $d(2) = d(1)^2$. The amount of temporary impact that has decayed after one period in this case is $1 - e^{-0.5 \cdot 1} = 0.3935$. After two periods the amount of temporary impact that has decayed is $1 - e^{-0.5 \cdot 2} = 0.632$. Figure 4.3d illustrates the quantity of temporary impact remaining for this function over several trade periods. Readers can verify that values in this figure match values computed above.

Example—Temporary Decay Formulation

The current price is \$30.00 and the temporary impact of each trade x_k is $f(x_k)$ (we exclude permanent impact here for simplicity). If the decay function parameter is γ the prices for our sequence of trades is:

$$P_0 = 30.00$$

The price of the first trade P_1 is the initial price plus the impact of the trade:

$$P_1 = P_0 + f(x_1)$$